BayesINE6. Modelos lineales generalizados. Regresión logística

Intro a Estadística Bayesiana

DataLab

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ICMAT-CSIC

Contents

- Chapter objectives
- Motivating example
- Computational problems with Bayesian analysis of logistic regression models
- Numerical solution in low dimension
- Solution based on asymptotic normality
- Final comments
- References
- Lab6

Chapter objectives

Objectives

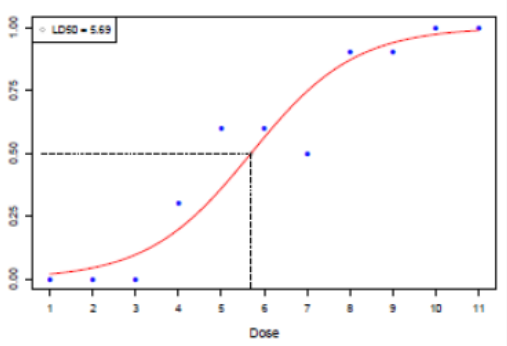
- Identify the computational problems that the Bayesian analysis of logistic regression bring to the fore.
- Describe a solution when the dimension is low.
- Describe a solution based on asymptotics.
- A lab illustrates concepts numerically

Motivating example

Motivation

Problem: Inference and prediction with logistic regression problems as example of GLMs

Typical example: *New anti-cancer drug being developed. Before human testing can begin, animal studies are needed to determine safe dosages. A dose-response experiment is carried out: 11 groups of 10 mice are treated with an increasing dose of drug and proportion of deaths are observed.*



Check `https://en.wikipedia.org/wiki/Logistic_regression`

Computational problems with Bayesian analysis of logistic regression models

Model formulation

Consider k dose levels $x_1, \ldots, x_k$. At dose x_i, observe n_i individuals with y_i successes.

Assume

$$Y_i \sim \text{Binomial}(n_i, p_i), \quad i = 1, \ldots, k$$

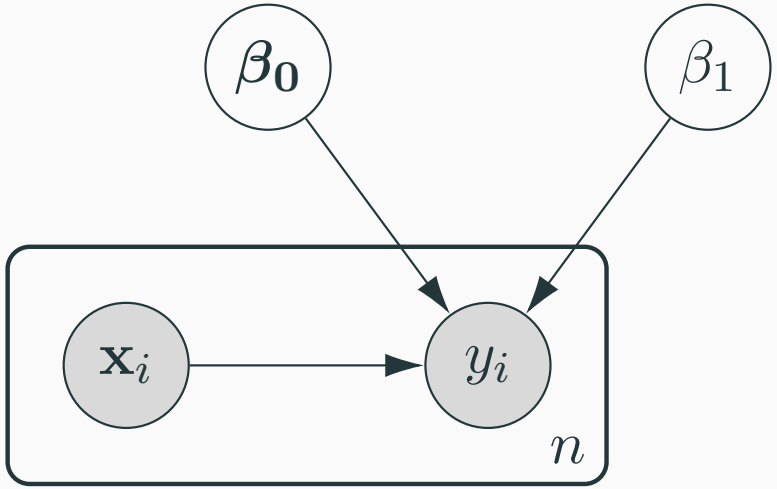
with logistic link:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 x_i$$

Equivalently:

$$p_i = \frac{1}{1 + \exp\left(-(\beta_0 + \beta_1 x_i)\right)}$$

Probabilistic setup



Likelihood

The likelihood function is:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{k} \binom{n_i}{y_i} p_i^{y_i} (1 - p_i)^{n_i - y_i}$$

Substituting p_i:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{k} \binom{n_i}{y_i} \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{n_i - y_i}$$

Log-likelihood:

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^{k} \left[y_i(\beta_0 + \beta_1 x_i) - n_i \log \left(1 + e^{\beta_0 + \beta_1 x_i} \right) \right] + \text{const}$$

MLE

MLE $(\hat{\beta}_0, \hat{\beta}_1)$ satisfies:

$$\frac{\partial \ell}{\partial \beta_0} = \sum_{i=1}^{k} (y_i - n_i p_i) = 0$$

$$\frac{\partial \ell}{\partial \beta_1} = \sum_{i=1}^{k} x_i (y_i - n_i p_i) = 0$$

Nonlinear. Solve numerically .

Bayesian analysis

Assume prior $\begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix}\mu_0\\ \mu_1\end{pmatrix}, \Sigma\right)$

Common choice is $\beta_0, \beta_1 \sim \mathcal{N}(0, \sigma^2)$ independent

Posterior is proportional to $p(\beta_0, \beta_1 \mid \mathbf{y}) \propto L(\beta_0, \beta_1) \cdot p(\beta_0, \beta_1)$

Thus

$$p(\beta_0, \beta_1 \mid \mathbf{y}) \propto \exp\left\{\sum_{i=1}^{k}\left[y_i(\beta_0 + \beta_1 x_i) - n_i \log(1 + e^{\beta_0 + \beta_1 x_i})\right]\right\} \times$$

$$\exp\left\{-\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{\beta} - \boldsymbol{\mu})\right\}$$

The posterior does not have a closed form!!!

Numerical solution in low dimension

Sampling from a piecewise constant approximation

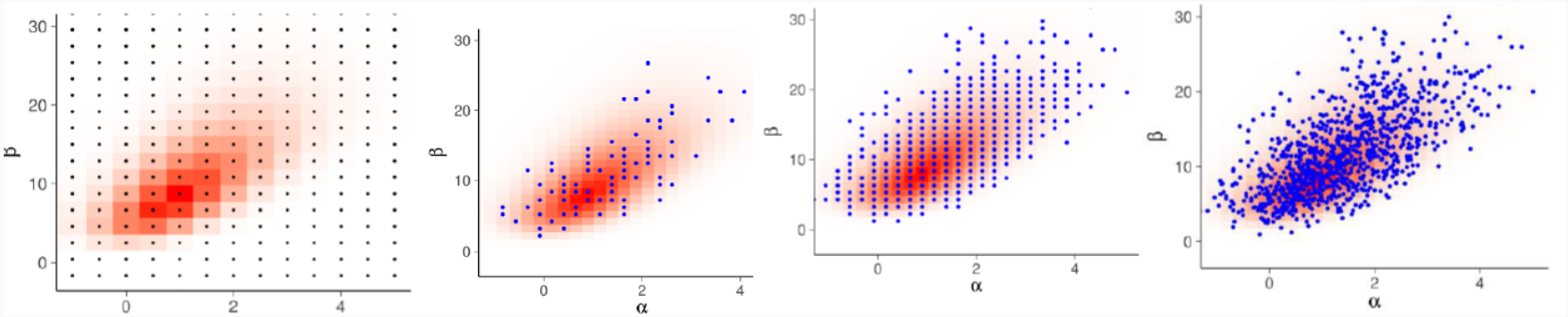
Sampling from the joint posterior through a piecewise constant approximation

- Obtain a finite region where posterior is concentrated.
- Choose grid over region
- Evaluate density over grid
- (Normalised) Density x Cell area approximates cell prob at each cell

Sample according to grid cell probabilities
Jitter for visualisation

Sampling from a piecewise constant approximation

Sampling from the joint posterior approximately



Sampling from a piecewise constant approximation

Posteriors of derived quantities.

The case of LD50, dose level at which death probability is 0.5

$$E\left(\frac{y}{n}\right) = \theta = logit^{-1}(\beta_0 + \beta_1 x) = 0.5$$

$$\beta_0 + \beta_1 x = logit(0.5) = 0$$

$$x_{LD50} = -\frac{\beta_0}{\beta_1}$$

Sample $(\beta_0^i, \beta_1^i) \sim p(\beta_0, \beta_1 | y, n, x)$

Approximate $E(x_{LD50}|y, n, x) \approx \frac{1}{n}\sum_i^n \frac{\beta_0^i}{\beta_1^i}$

Solution based on asymptotic normality

Asymptotics: discrete case

Assume parameter space is discrete and there is a true underlying distribution.

For the given parametric model, consider the parameter minimising Kullback-Leibler divergence to the true distribution.

If minimiser has positive prior probability, the posterior concentrates over the minimiser as data accumulates.

Asymptotics: continuous case

If parameter space is compact and A is a neigborhood of minimiser with positive prior probability,
the posterior concentrates in A as data accumulates.

Asymptotics: normality with posterior mode

Under certain regularity conditions, the posterior approaches normality around the posterior mode as data accumulates

- Parameter space in R^p
- Mode unique
- Mode interior to parameter space
- Density continuous and positive at mode
- Log-likelihood twice differentiable at neighbourhood of mode

Asymptotics: normality with MLE

- As sample size increases, likelihood dominates (a non-degenerate) prior
- Prior behaves as approximately constant
- Posterior mode approximated by MLE
- Inverse Fisher's info matrix at MLE approximates variance

Inverse Fisher's info at MLE

Log-likelihood $\ell(\theta) = \log L(\theta)$ and $\hat{\theta}$ the MLE

Fisher information matrix at MLE $\mathcal{I}(\hat{\theta}) = -\mathbb{E}\left[\left.\frac{\partial^2 \ell(\theta)}{\partial\theta\,\partial\theta^\top}\right|_{\theta=\hat{\theta}}\right]$.

Inverse Fisher info approximates the covariance matrix of MLE

$\mathrm{Var}(\hat{\theta}) \approx \mathcal{I}(\hat{\theta})^{-1}$

Observed Fisher info $\mathcal{J}(\hat{\theta}) = -\left.\frac{\partial^2 \ell(\theta)}{\partial\theta\,\partial\theta^\top}\right|_{\theta=\hat{\theta}}$.

Usage

Approximate posterior by normal

Moments and regions as in normal

Final comments

Final comments

- Non-availability of posterior is the norm
- Normal asymptotics is too demanding for many cases, thus requiring other strategies
- Logistic regression as a model for classification (supervised learning with discrete response)

References

References

Gelman, Carlin, et al (2014) Bayesian Data Analysis.

French, Ríos Insua (2000) Statistical Decision Theory.

Lab6

Lab6

Both strategies showed with logistic regression